LAB 7: Fuzzing for Input Validation Bugs

Screenshot document

NAME	SRN	SECTION
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Original_application

```
$ python lab7/coupon_validator.py
==== Running split_code ====
Enter a coupon code (e.g. save-20): save-20
Prefix and Value: ('SAVE', 20)

==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): 199,49.5,320
Most Expensive Item: 320.0

==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: 2025-06-15
Days Since 2024: 530
```

Failing fuzz tests

```
$ python -m pytest lab7/test_coupon_validator.py -v
==================== test session starts ====================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\SE\lsb6-7\L6_L7_31_coupon_validator
plugins: anyio-4.14.2, hypothesis-6.170.15
collected 3 items

lab7/test_coupon_validator.py::test_split_code_no_crash FAILED [ 33%]
lab7/test_coupon_validator.py::test_most_expensive_no_crash FAILED [ 66%]
lab7/test_coupon_validator.py::test_expiry_days_no_crash FAILED [100%]

========================= FAILURES =========================
_____________ test_split_code_no_crash _____________

    @given(st.one_of(st.text(), st.none()))
>   def test_split_code_no_crash(s):
                ^^^

lab7\test_coupon_validator.py:8:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
lab7\test_coupon_validator.py:10: in test_split_code_no_crash
    split_code(s)
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

code = None

    def split_code(code):
>       prefix, value = code.strip().upper().split('-')
                        ^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'
E       Failing test case: test_split_code_no_crash(
E           s=None,
E       )
```

```
lab7\coupon_validator.py:2: AttributeError
_____________ test_most_expensive_no_crash _____________

    @given(st.one_of(st.text(), st.none()))
>   def test_most_expensive_no_crash(s):
                  ^^^

lab7\test_coupon_validator.py:16:

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
lab7\test_coupon_validator.py:18: in test_most_expensive_no_crash
    most_expensive(s)
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

prices_csv = None

    def most_expensive(prices_csv):
>       prices = [float(p) for p in prices_csv.split(',')]
                                    ^^^^^^^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'split'
E       Failing test case: test_most_expensive_no_crash(
E           s=None,
E       )

lab7\coupon_validator.py:6: AttributeError
_____________ test_expiry_days_no_crash _____________

    @given(st.one_of(st.text(), st.none()))
>   def test_expiry_days_no_crash(s):
                  ^^^

lab7\test_coupon_validator.py:24:

_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
lab7\test_coupon_validator.py:26: in test_expiry_days_no_crash
    expiry_days(s)
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _
```

```
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

date_text = None

    def expiry_days(date_text):
>       year, month, day = [int(p) for p in date_text.split('-')]
                                    ^^^^^^^^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'split'
E       Failing test case: test_expiry_days_no_crash(
E           s=None,
E       )

lab7\coupon_validator.py:10: AttributeError
==================== warnings summary ====================
..\..\..\Users\prart\AppData\Roaming\Python\Python311\site-packages\anyio\pytest_plugin.py:14
  C:\Users\prart\AppData\Roaming\Python\Python311\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
================= short test summary info =================
FAILED lab7/test_coupon_validator.py::test_split_code_no_crash - AttributeError: 'NoneType' object has no attribute 'strip'
FAILED lab7/test_coupon_validator.py::test_most_expensive_no_crash - AttributeError: 'NoneType' object has no attribute 'split'
FAILED lab7/test_coupon_validator.py::test_expiry_days_no_crash - AttributeError: 'NoneType' object has no attribute 'split'
=============== 3 failed, 1 warning in 10.38s ===============
```

Fixed application output

```
$ python lab7/coupon_validator.py
==== Running split_code ====
Enter a coupon code (e.g. save-20): save-20
Prefix and Value: ('SAVE', 20)

==== Running most_expensive ====
Enter item prices as CSV (e.g. 199,49.5,320): 199,49.5,320
Most Expensive Item: 320.0

==== Running expiry_days ====
Enter an expiry date as YYYY-MM-DD: 2025-06-15
Days Since 2024: 531
```

Execution of Hypothesis tests after fixing input validation bugs.

```
prart@Inola MINGW64 /c/SE/lsb6-7/L6_L7_31_coupon_validator (main)
$ python -m pytest lab7/test_coupon_validator.py -v
========== test session starts ==========
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0 -- C:\Program Files\Python311\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\SE\lsb6-7\L6_L7_31_coupon_validator
plugins: anyio-4.14.2, hypothesis-6.170.15
collected 3 items

lab7/test_coupon_validator.py::test_split_code_no_crash PASSED [ 33%]
lab7/test_coupon_validator.py::test_most_expensive_no_crash PASSED [ 66%]
lab7/test_coupon_validator.py::test_expiry_days_no_crash PASSED [100%]

========== warnings summary ==========
..\..\..\Users\prart\AppData\Roaming\Python\Python311\site-packages\anyio\pytest_plugin.py:14
  C:\Users\prart\AppData\Roaming\Python\Python311\site-packages\anyio\pytest_plugin.py:14: PytestRemovedIn10Warning: _pytest.python.CallSpec2 has been renamed to CallSpec.
  The CallSpec2 alias will be removed in pytest 10.
  Update imports to use CallSpec instead.
  See https://docs.pytest.org/en/stable/deprecations.html#callspec2-renamed
    from _pytest.python import CallSpec2

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
========== 3 passed, 1 warning in 2.52s ==========
```